

CareerPilot AI – Complete Project Specification

Project Overview

CareerPilot AI is an AI-powered career management platform that helps users:

- Manage job applications
- Track application progress
- Upload and analyze resumes
- Calculate ATS compatibility scores
- Match resumes with job descriptions
- Identify missing skills
- Generate AI-based resume improvement suggestions
- Prepare for interviews
- Visualize career analytics and progress

The goal is to provide a complete career-assistance platform in a single application.

Technology Stack

Frontend

- React.js
- React Router
- Axios
- Context API
- Tailwind CSS / Material UI

Backend

- Node.js
- Express.js
- JWT Authentication

Database

- MongoDB
- Mongoose

AI Features

- OpenAI API / Gemini API
- PDF Parsing

- Resume Analysis

File Handling

- Multer
- PDF Parse

Deployment

- Frontend: Vercel
 - Backend: Render/Railway
 - Database: MongoDB Atlas
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MODULE 1: Authentication System

Purpose

Provide secure user registration and login.

Features

User Registration

Fields:

- Name
- Email
- Password

API:

POST /api/auth/register

User Login

Fields:

- Email
- Password

API:

POST /api/auth/login

JWT Authentication

Generate token after login.

Protected Routes

Users can only access their own data.

User Profile

API:

GET /api/auth/profile

Returns:

- User Name
- Email
- Account Creation Date

Status: COMPLETED

MODULE 2: Job Application Tracker

Purpose

Track all job applications.

Application Fields

- Company Name
- Job Role
- Status
- Application Date
- Notes
- User Reference

Status Types

- Applied
- Interview
- Rejected
- Offer
- Accepted

Features

Create Application

POST /api/applications

View Applications

GET /api/applications

Update Application

PUT /api/applications/:id

Delete Application

DELETE /api/applications/:id

Status: COMPLETED

MODULE 3: Search, Filter & Pagination

Purpose

Manage large application lists.

Search

Search by:

- Company Name
- Job Role

Filter

Filter by:

- Applied
- Interview
- Rejected
- Offer

Pagination

Example:

- Page 1
- Page 2
- Page 3

Status: COMPLETED

MODULE 4: Dashboard Analytics

Purpose

Provide quick statistics.

Dashboard Cards

- Total Applications
- Applied Count
- Interview Count
- Rejected Count
- Offer Count

API

GET /api/applications/stats

Status: COMPLETED

MODULE 5: Resume Management

Purpose

Store and manage resumes.

Resume Upload

Upload PDF Resume.

API

POST /api/resume/upload

Resume Storage

Store:

- File Name
- Upload Date
- File Path
- User ID

Resume History

Users can see previously uploaded resumes.

Status: PENDING

MODULE 6: Resume Parsing

Purpose

Extract information from resumes.

Extracted Information

- Skills
- Education
- Experience
- Projects
- Certifications

Process

Upload PDF ↓ Parse PDF ↓ Extract Text ↓ Store Text

Status: PENDING

MODULE 7: ATS Score Analyzer

Purpose

Evaluate resume quality.

Output

ATS Score

Example:

82 / 100

Evaluation Criteria

- Keyword Presence
- Resume Structure
- Skills Coverage
- Experience Section
- Education Section

Output Example

ATS Score: 82

Strengths:

- Good React Experience
- Strong Projects

Weaknesses:

- Missing Cloud Skills

Status: PENDING

MODULE 8: Job Description Matching

Purpose

Compare resume with job description.

Inputs

- Resume
- Job Description

Output

Match Percentage

Example:

87%

Missing Keywords

Example:

Required:

- React
- Node.js
- AWS
- Docker

Found:

- React
- Node.js

Missing:

- AWS
- Docker

Status: PENDING

MODULE 9: Skill Gap Analysis

Purpose

Identify skills missing for target jobs.

Features

Analyze:

- Resume Skills
- Job Description Skills

Output

Recommended Skills:

- Docker

- AWS
- CI/CD
- Kubernetes

Status: PENDING

MODULE 10: AI Resume Review

Purpose

Provide personalized resume feedback.

Output Sections

Strengths

Weaknesses

Suggestions

Example

Strengths:

- Strong frontend experience

Weaknesses:

- No quantified achievements

Suggestions:

- Add measurable project results
- Include cloud projects

Status: PENDING

MODULE 11: Interview Preparation

Purpose

Generate interview questions.

Input

Target Role

Example:

Frontend Developer

Generated Questions

Technical Questions

Behavioral Questions

HR Questions

Advanced Version

Generate questions directly from:

- Resume
- Job Description

Status: PENDING

MODULE 12: Career Analytics

Purpose

Provide career insights.

Metrics

Most Applied Role

Interview Success Rate

Offer Rate

Rejection Rate

Average Applications Per Month

Most Common Missing Skills

Status: PENDING

MODULE 13: Data Visualization

Purpose

Display progress visually.

Charts

Application Trend

Monthly Applications

Status Distribution

Interview Success Rate

Libraries

- Recharts or
- Chart.js

Status: PENDING

MODULE 14: User Profile

Purpose

Manage account information.

Features

View Profile

Update Profile

Change Password

Delete Account

Status: PENDING

MODULE 15: Notifications (Optional)

Purpose

Keep users informed.

Notifications

Interview Reminder

Application Follow-up Reminder

Resume Improvement Suggestions

Status: FUTURE ENHANCEMENT

MODULE 16: Admin Dashboard (Optional)

Purpose

Manage platform usage.

Features

Total Users

Total Resumes

Total Applications

System Analytics

Status: FUTURE ENHANCEMENT

Frontend Pages

Authentication

- Login Page
- Register Page

Application Management

- Dashboard
- Applications

Resume

- Resume Upload
- Resume Analysis

Interview

- Interview Preparation

Profile

- User Profile

Analytics

- Career Insights

Settings

- Account Settings

Final Project Completion Checklist

Backend

[✓] Authentication

[✓] Application CRUD

[✓] Search

[✓] Filter

☒ Pagination

☒ Dashboard Analytics

☐ Resume Upload

☐ Resume Parsing

☐ ATS Analyzer

☐ Job Match Engine

☐ Skill Gap Analysis

☐ Resume Review

☐ Interview Generator

Frontend

☐ Login Page

☐ Register Page

☐ Dashboard Page

☐ Applications Page

☐ Resume Upload Page

☐ Resume Analysis Page

☐ Interview Page

☐ Profile Page

Analytics

☐ Charts

☐ Career Insights

Deployment

☐ Backend Deployment

☐ Frontend Deployment

☐ MongoDB Atlas

Project Status Target: 100% Complete